



*Faculty of Engineering & Technology Electrical & Computer
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ENCS4380

HW1 - Intelligent Flood Detection System (ISI): Smart Street Lamp Post

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Sec.: 1

Date: 8/10/2023

1. The concept

The combination of intelligent sensor systems and disaster management has given rise to creative solutions in the fast-changing environment of smart technology. The Smart Street Lamp Post, a project made by students enrolled in the Software of Places program at PUC-Rio University and published on www.softwareofplaces.com, is one such example. This cutting-edge program demonstrates the power of Intelligent Sensor Systems (ISI) to transform urban infrastructure.

This is what it looks like in the pictures below:

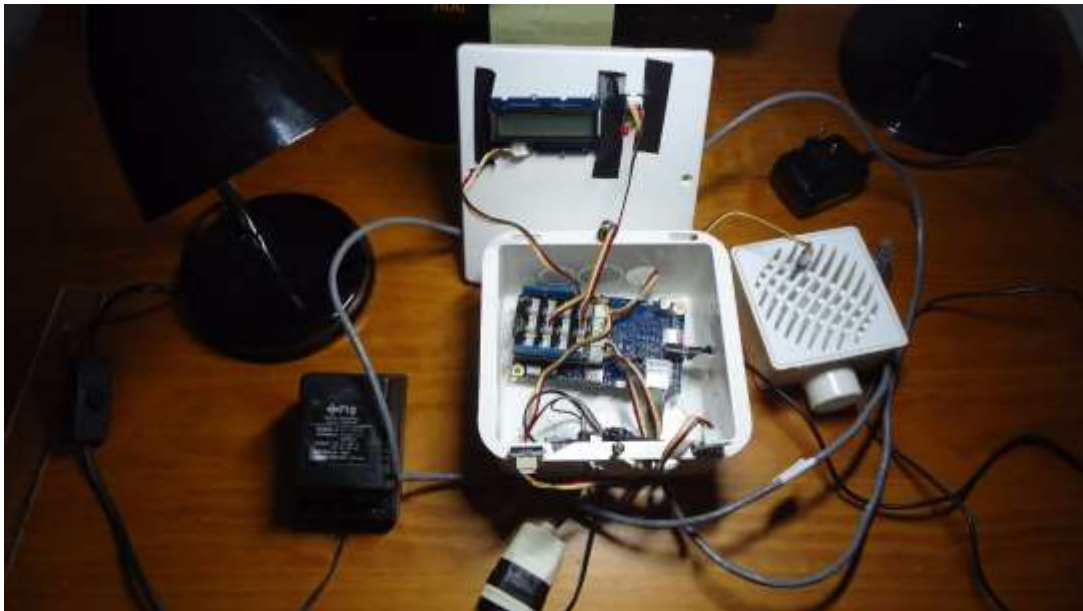


Figure 1&2: Smart Streep Lamp Post



2. Sensors

The system employed an array of sensors including Grove Moisture Sensor, Grove Temperature Sensor, and Grove Light Sensor. The Grove Moisture Sensor, placed inside a drain, detects moisture content and water presence, crucial for flood prediction. The Grove Temperature Sensor assesses ambient temperature, while the Grove Light Sensor detects light levels, facilitating intelligent lamp control.

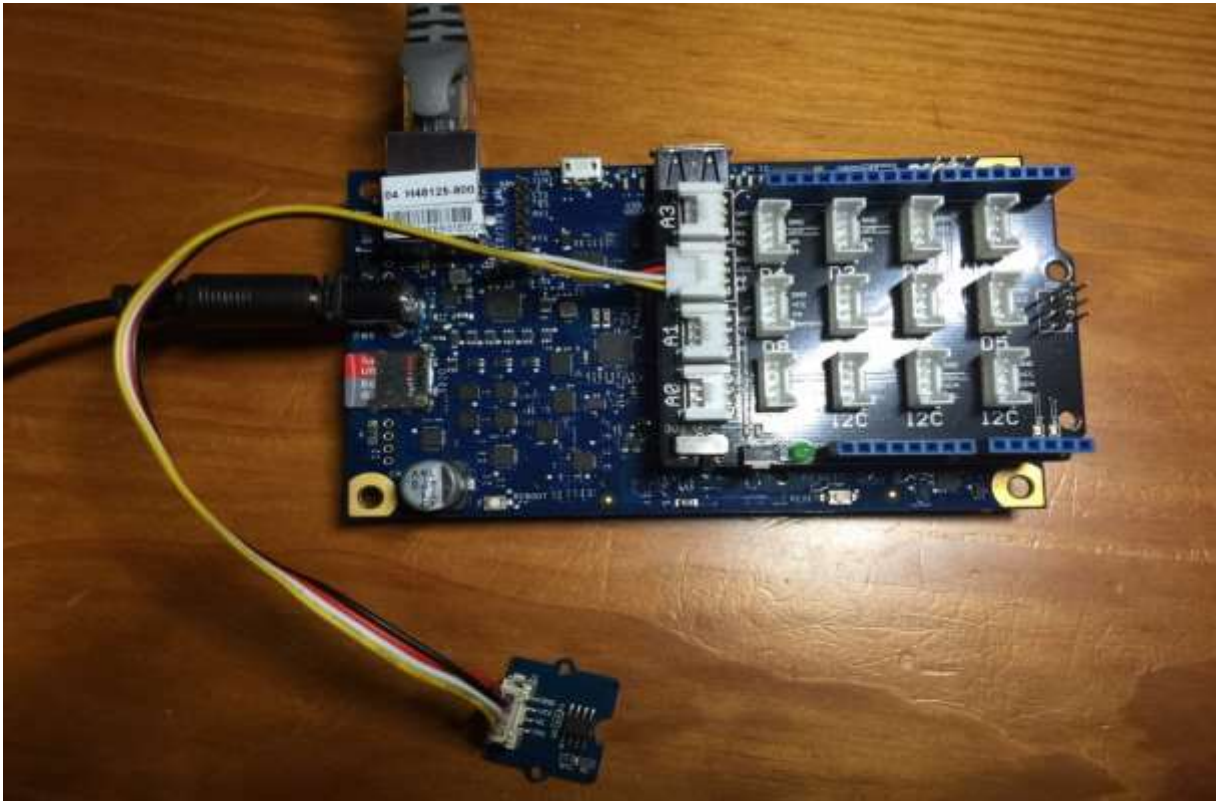


Figure 3: Initial Setup and Temperature Sensor

3. Microcontrollers:

Driving the system's computational prowess is the Intel Galileo Gen 2 microcontroller. Its role is pivotal, overseeing data processing, algorithm execution, and actuator control. The project leverages the Cylon.js framework, a versatile JavaScript framework, enabling seamless communication and data integration. This framework unifies diverse sensors and microcontrollers, embodying the essence of ISI through its adaptability and compatibility.

4. Actuators:

To produce visual alerts for drivers, the project used Grove LCD RGB screens and Grove LEDs. The Grove Relay controlled the street lamp, allowing it to turn on and off automatically. Servo motors were used to modify the light post's height, keeping it above the water level during flooding and thereby improving safety.



Figure4: servo motor



Figure5: Grove LED



Figure6: Grove LCD RGB



Figure7: Flood Detection and Road Display Board

5. Algorithm where Intelligence is utilized:

The flood detection algorithm, a pivotal component, relies on predictive analysis. By comparing sensor readings against a predefined THRESHOLD value, the algorithm identifies imminent flood states. The intelligent processing involves buffered values, ensuring accuracy and preventing false alarms. This algorithm exemplifies the intelligent sensing paradigm, where raw sensor data is transformed into actionable insights.

- The THRESHOLD value signifies the moisture sensor's reading when the drains are on the verge of being filled with water. This value is crucial as it indicates an imminent flood situation, where the water volume entering the drain matches or exceeds its output flow rate. The processBuffer method compares this THRESHOLD value to the average of the last MAX_SIZE values obtained from the moisture sensor, ensuring accuracy in flood predictions.

When the system identifies an impending flood, timely notifications are essential for road safety. To create a visual alert system, the project utilizes the Grove LCD display and Grove LED. In such situations, the LCD display will show "FLOOD RISK," and the LED will start flickering, warning drivers about the hazardous road condition. Under normal circumstances, the LCD will display "Road open," and the LED will remain switched off.

6. Layout and Assembly:

The physical assembly of the Smart Street Lamp Post involves meticulous design and 3D printing technology. Custom mounts for sensors are created using 3D printing, ensuring stability and optimal functionality. Sensors are strategically placed, maximizing coverage while minimizing vulnerability to environmental elements. The layout emphasizes seamless integration, enabling efficient data flow between sensors, microcontroller, and actuators.

7. System Evaluation:

Extensive system assessment tests were carried out to evaluate performance, dependability, and responsiveness. Simulated flood scenarios were used to evaluate the accuracy of flood predictions as well as the efficacy of visual alerts. The reaction speed, energy economy, and real-time data processing capabilities of the system were carefully assessed. The findings demonstrated the system's durability and dependability, verifying its efficacy in real-world crisis management scenarios.

Conclusion:

Finally, the Smart Street Lamp Post project serves as a model application of the Intelligent Sensor System (ISI) idea. The system highlights the transformational potential of ISI in disaster management by seamlessly integrating modern sensors, microcontrollers, actuators, and intelligent algorithms. Students' joint efforts and inventive use of technology highlight the significance of intelligent sensing in designing resilient and proactive urban landscapes. This initiative not only improves catastrophe preparedness, but it also demonstrates the ability of intelligent technology in alleviating real-world issues, therefore creating a precedent for future urban infrastructure advances.